PRAISING HARRIS

RED TEAM / OFFENSIVE SECURITY / AI RED TEAMING

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2 Merged Tier-1 PRs SQLGlot & LiteLLM · **Self-assessed CVSS 7.5** LiteLLM BOLA found, fix PR open · **100% / 0% FPR** ZeroTrace syscall implant hunter · **Top 2%** TryHackMe, global

PROFILE

Offensive-security undergraduate available for a 6-month full-time industrial co-op (Nov 14, 2026 – Summer 2026 with university NOC clearance and credit eligibility), finding authorization vulnerabilities in production AI infrastructure (LiteLLM, FastMCP), submitting fixes upstream, implementing published LLM jailbreak attacks (PAIR, TAP, Crescendo, GCG), and researching weight/pickle supply-chain backdoors.

EDUCATION

Indian Institute of Information Technology, Kottayam 2024 – 2028 (expected)
B.Tech Computer Science & Engineering (Cybersecurity) | CGPA 7.9
6-Month Co-op Availability: Nov 14, 2026 – Summer 2026 | University NOC Approved & Academic Credits Eligible

EXPERIENCE

VAPT Intern (Remote), Cyberops Jul 2026 – Oct 2026

- Completed company-assigned vulnerability assessment and penetration testing training covering Burp Suite, Nmap and the OWASP Top 10, practised on TryHackMe and HackTheBox machines.

PROJECTS

Vulnerability Research & Upstream Protocol Patches: LiteLLM, FastMCP, SQLGlot github.com/Pyhroff
BerriAI/litellm (#47105, #47312) · PrefectHQ/fastmcp (#6063) · Anthropic MCP SDK · SQLGlot (Merged #8780)

- Discovered a Broken Object-Level Authorization (BOLA) bypass in BerriAI/litellm (self-assessed CVSS 7.5, PR #47105): tokens and teams with access_group_ids and empty direct models triggered wildcard fallback, granting unrestricted model access across enterprise endpoints; authored upstream fix and unit tests; scoped /v1/mcp/access_groups to caller accessible servers (PR #47312).
- PrefectHQ/fastmcp (PR #6063): Uncovered host event loop starvation and sandbox escape (CWE-653 / CWE-400) in MontySandboxProvider and CodeMode; patched with atomic wall-clock timeout and external callback limit (65/65 tests pass); hardened Anthropic MCP SDK against RFC 6749 §5.2 OAuth violations (PRs #3764, #3765, 175 tests); shipped merged AST fix in SQLGlot (#8780).

ZeroTrace: x86_64 Direct/Indirect Syscall Implant Hunter github.com/Pyhroff/zerotrace
Python · x86_64 Syscall Stubs · Shannon Entropy · Graph Min-Cut · 10/10 pytest pass

- Syscall Implant Hunter: Engineered direct and indirect syscall forensics detecting Hell's Gate, Halo's Gate, Tartarus' Gate, and SysWhispers3 stubs jumping directly into ntdll.dll; implemented disk-vs-memory .text byte diffing and sliding-window Shannon entropy over private RWX regions (100% recall, 0% FPR across 40 memory regions in 13.06ms).
- Active Directory & Evasion Analysis: Implemented BloodHound privilege graph Dijkstra shortest-path and Edmonds-Karp min-cut chokepoint analysis severing 100% Tier-0 attack paths (180 → 0); built 5-layer CLI deobfuscation engine evaluating Sysmon Event ID 8/10 unbacked call traces.

PromptStrike: LLM Red-Teaming Framework github.com/Pyhroff/promptstrike
Python · FastAPI · WebSocket · SQLite · GitHub Actions

- Implemented PAIR, TAP, Crescendo and GCG (white-box) jailbreak attacks from their papers, scored by a JailbreakBench-style 1–10 judge, with 50 behaviour goals mapped to the OWASP LLM Top 10.
- Ran multi-model sweeps across Groq Llama 3.3 70B, OpenAI and Ollama targets to compare attack success rate side by side; every attack tree is stored in SQLite and streamed live over WebSocket.
- Added a CI gate that fails a build when attack success rate exceeds a threshold, and a Defense Shield (10-rule input scanner plus output jailbreak classifier) that wraps any model adapter; 61 tests pass offline.

AdversaGen: Adaptive Attack-Payload Generator github.com/Pyhroff/adversagen
Python · SciPy · evolutionary search

- Evolved evasions (zero-width characters, homoglyphs, paraphrase, indirection) against a naive blocklist on four surfaces (prompt injection, tool poisoning, memory injection, RAG injection); 240 runs (30 seeds × 2 conditions × 4 surfaces).
- Found random mutation beat reflection-guided mutation on every surface (evasion 0.86–0.91 vs 0.63–0.72, $p < 0.0001$), traced it to lost population diversity, and reported the negative result rather than tuning it away.

redteam-toolkit: Authorized-Use Red Team Framework github.com/Pyhroff/redteam-toolkit

Python · about 6,700 lines across 22 modules

- Recon modules for port scanning, subdomain enumeration, directory fuzzing, and WAF and tech-stack fingerprinting mapped to CVEs.
- Web-testing modules for SQL injection, LFI with PHP wrappers and XSS, plus a Linux privilege-escalation audit using GTFOBins and an OPSEC risk advisor.
- An orchestrator chains the modules and logs findings per engagement.

ModelHawk: ML Model Backdoor Scanner github.com/Pyhroff/ModelHawk

Python (stdlib only) · pytest · GitHub Actions

- Built both sides of the pickle-deserialization attack on PyTorch models: a generator for harmless malicious samples whose payload runs on torch.load(), and a static scanner that reads the opcode stream (REDUCE / __reduce__ chains) without loading the model; 8 tests. Write-up published on Medium.

aisec-suite 0.3.0: MCP Tool-Poisoning Detection & Precision Study github.com/Pyhroff/aisec-suite

Python · Typer · AST taint analysis · SARIF · PyPI

- Built a scanner for MCP tool poisoning and over-broad tool scope, then measured it: 37.5% of v0.6 HIGH permission_scope findings were accurate across 90 public MCP repos, and the 0.3.0 structural detector reaches 48.1% held-out recall at 0.03% false positives on 3,974 real tool descriptions.

SKILLS

Offensive Burp Suite, Nmap, OWASP Top 10, SQLi / LFI / XSS testing, recon and OSINT, Linux privilege escalation, MITRE ATT&CK

AI red teaming PAIR, TAP, Crescendo, GCG, JailbreakBench, OWASP LLM Top 10, MITRE ATLAS, prompt / tool / memory / RAG injection, pickle RCE in ML model files

Engineering Python, JavaScript, SQL, Bash; FastAPI, pytest, GitHub Actions, Docker, SciPy

CERTIFICATIONS, WRITING & HACKATHONS

TryHackMe AI Security path (25 Jun 2026, ID THM-N1JNCY2I4B) and Pre-Security path; top 2% globally, 177 rooms completed. Fortinet: Network Security Expert Level 1 (NSE 1).

Medium six security articles, including “I Built a Malware Scanner for AI Models. Here’s What I Found”. CTFs: participated in several competitions.

Hackathons PSBs Hackathon 2026 (PhantomGrid, behavioural-biometric authentication built to resist replay and mimicry); USAII Global AI 2026 (Lucid, prompt-injection resistance as a design requirement).